Configure external-snapshot-metadata sidecar container for RBD deployment - Developer Preview OpenShift Data Foundation 4.20

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A new CSI API of OpenShift Container Platform enables identification of the changed blocks between snapshots of CSI volumes. These APIs help with efficient and incremental backups by allowing backup applications to retrieve only the data that has changed. For more information, see How to use Changed Block Tracking in OpenShift 4.20.

To support this feature, Ceph-CSI needs to include an `external-snapshot-metadata` sidecar in the RBD controller plugin.

This document outlines how the Ceph-CSI Operator manages the deployment of the RBD controller plugin with the `external-snapshot-metadata` sidecar.

Note: Only the RBD driver supports the SnapshotMetadata capability.

Deploying RBD controller plugin with the external-snapshot-metadata sidecar

1. Enable the feature gate to install the `SnapshotMetadataService` CRD.

```
$ oc edit featuregates.config.openshift.io cluster

```
spec:
  customNoUpgrade:
    enabled:
    - ExternalSnapshotMetadata
  featureSet: CustomNoUpgrade
```
```

a. Verify that CRD is installed in the cluster.
~~~
$ oc get crd | grep -i snapshotmetadataservice
snapshotmetadataservices.cbt.storage.k8s.io 2025-09-02T08:51:40Z
~~~

2. Create a service for the Ceph-CSI RBD deployment pod.

```
$ cat <<EOF | oc create -f -
apiVersion: v1
kind: Service
metadata:
  name: rbd-csi-ceph-com-metadata
  namespace: openshift-storage
spec:
  ports:
  - name: snapshot-metadata-port
    port: 6443
    protocol: TCP
    targetPort: 50051
  selector:
    app: openshift-storage.rbd.csi.ceph.com-ctrlplugin  # RBD controller plugin pod
label
  type: ClusterIP
EOF
```

3. Provision certs and secrets.

   - Generate TLS certificates using your preferred method (for example, `cert-manager`). The certificates must be valid for the service domain created in the previous step: **rbd-csi-ceph-com-metadata.openshift-storage**

- Create a TLS secret with the generated certificates:

  ~~~
  oc create secret tls snapshot-metadata-tls \
  --namespace=openshift-storage \
  --cert=server-cert.pem \
  --key=server-key.pem
  ~~~

Or, optionally, use the OCP Service Certificates. For more information, see Add a service certificate in OpenShift documentation.

```
$ oc -n openshift-storage annotate service rbd-csi-ceph-com-metadata
service.beta.openshift.io/serving-cert-secret-name=snapshot-metadata-tls

# For the CA Bundle Create ca-bundle configmap with
service.beta.openshift.io/inject-cabundle=true annotation

$ cat <<EOF | oc create -f -
apiVersion: v1
kind: ConfigMap
metadata:
  name: ca-bundle
  namespace: openshift-storage
data:
EOF
configmap/ca-bundle created


$ oc -n openshift-storage annotate cm ca-bundle service.beta.openshift.io/inject-
cabundle=true
configmap/ca-bundle annotate

$ oc -n openshift-storage get cm ca-bundle -oyaml
apiVersion: v1
data:
  service-ca.crt: |
    -----BEGIN CERTIFICATE-----
    ...
    -----END CERTIFICATE-----
kind: ConfigMap
...

# get the base64 of the CA bundle
$ oc get cm -n openshift-storage ca-bundle -ojsonpath="{.data.service-ca\.crt}" |
base64
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSURVVENDQWptZ0F3SUJBZ0lJRGp....
```

4. Create a `SnapshotMetadataService` CR for the RBD driver.
   Ensure that the name of this CR matches the RBD driver CR name.
   - address: Should point to the service created in step 2, replace , , and with your actual values from the earlier step.
   - audience: Recommended to use the CSI driver name for consistency
   - caCert: Base64-encoded CA certificate bundle (If you have used OpenShift certificates then the CA bundle should be available in the ConfigMap ca-bundle created in the previous step)

```
$ cat <<EOF | oc create -f -
apiVersion: cbt.storage.k8s.io/v1alpha1
kind: SnapshotMetadataService
metadata:
  name: openshift-storage.rbd.csi.ceph.com
spec:
  address: rbd-csi-ceph-com-metadata.openshift-storage:6443
  audience: openshift-storage.rbd.csi.ceph.com
  caCert: <ca-bundle>
EOF
```

5. Provide the TLS secret required for the `external-snapshot-metadata` sidecar as a volume mount in the RBD driver CR.

```
$ oc -n openshift-storage edit driver openshift-storage.rbd.csi.ceph.com
apiVersion: csi.ceph.io/v1
kind: Driver
metadata:
  name: openshift-storage.rbd.csi.ceph.com
  namespace: openshift-storage
spec:
  controllerPlugin:
    volumes:
    - mount:
        mountPath: /tmp/certificates  # Must be /tmp/certificates - required by
sidecar
        name: tls-key
      volume:
        name: tls-key # Must be "tls-key"
        secret:
          secretName: snapshot-metadata-tls  # The TLS secret name created in Step
3
```

6. Verify that the `csi-snapshot-metadata` container has started in RBD controller plugin pod.

```
$ oc logs openshift-storage.rbd.csi.ceph.com-ctrlplugin-846cdbd94d-ss5pk csi-
snapshot-metadata
I1028 07:47:01.360523       1 sidecar.go:80] Version: v0.1.0
I1028 07:47:01.361215       1 envvar.go:172] "Feature gate default state"
feature="InformerResourceVersion" enabled=false
I1028 07:47:01.361235       1 envvar.go:172] "Feature gate default state"
feature="WatchListClient" enabled=false
I1028 07:47:01.361241       1 envvar.go:172] "Feature gate default state"
feature="ClientsAllowCBOR" enabled=false
I1028 07:47:01.361247       1 envvar.go:172] "Feature gate default state"
feature="ClientsPreferCBOR" enabled=false
I1028 07:47:01.362563       1 connection.go:234] "Connecting"
address="unix:///csi/csi.sock"
I1028 07:47:01.364699       1 connection.go:264] "GRPC call"
method="/csi.v1.Identity/GetPluginInfo" request="{}"
I1028 07:47:01.366762       1 connection.go:270] "GRPC response" response="
{\"name\":\"openshift-storage.rbd.csi.ceph.com\",\"vendor_version\":\"release-
4.20\"}" err=null
I1028 07:47:01.366813       1 sidecar.go:88] CSI driver name: "openshift-
storage.rbd.csi.ceph.com"
I1028 07:47:01.367767       1 sidecar.go:273] GRPC server started listening on port
50051
I1028 07:47:01.367828       1 common.go:143] "Probing CSI driver for readiness"
I1028 07:47:01.367931       1 connection.go:264] "GRPC call"
method="/csi.v1.Identity/Probe" request="{}"
I1028 07:47:01.368694       1 connection.go:270] "GRPC response" response="{}"
err=null
I1028 07:47:01.368752       1 connection.go:264] "GRPC call"
method="/csi.v1.Identity/GetPluginCapabilities" request="{}"
I1028 07:47:01.369611       1 connection.go:270] "GRPC response" response="
{\"capabilities\":[{\"service\":{\"type\":\"CONTROLLER_SERVICE\"}},
{\"volume_expansion\":{\"type\":\"ONLINE\"}},{\"service\":
{\"type\":\"VOLUME_ACCESSIBILITY_CONSTRAINTS\"}},{\"service\":
{\"type\":\"GROUP_CONTROLLER_SERVICE\"}},{\"service\":
{\"type\":\"SNAPSHOT_METADATA_SERVICE\"}}]}" err=null
I1028 07:47:01.369672       1 health.go:48] Setting status to SERVING
```

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